

Consumer Segmentation & Strategic Analysis Report

Advanced K-Means Clustering and Chi-Square Independence Analysis

Dataset Source:
Rural_vs_Urban_Consumer_Survey_EN_updated

Algorithm: K-Means Clustering (Seed: 25)

Total Respondents: 149

Analysis Type: Cluster Analytics & Chi-Square Independence Tests

1. Executive Summary

This report outlines the strategic data restructuring and statistical analysis performed on the *Rural vs. Urban Consumer Survey* dataset. To extract actionable consumer personas, five categorical columns ("Location Type", "Gender", "Current Mobile Phone Brand", "Factors For Buying", and "Preferred Purchase Channel") were excluded during the initial phase to perform unsupervised machine learning on five core numerical metrics.

A 3-cluster K-Means algorithm (random seed 25) was executed on standardized features (utilizing a `StandardScaler` to eliminate dimensional differences across age, income, and spending ranges). Following cluster generation, a series of Chi-Square Independence Tests were conducted to evaluate how strongly the excluded demographic and behavioral variables correlate with the resulting clusters. The results reveal a highly robust demographic split predominantly driven by economic geography (Urban vs. Rural) and brand ecosystems.

2. K-Means Clustering Architecture & Centroids

The clustering algorithm effectively mapped the 149 respondents into three distinctive socio-economic brackets based on five numerical metrics. Missing values (such as an empty field in Age) were successfully imputed using the column median to preserve data integrity before scaling.

Final Centroid Matrix (Converted back to original units)

Cluster ID	Age (Years)	Monthly Household Income (₹)	Amount Spent on Previous Phone (₹)	Usage Before Replacing (Months)	Research Before Purchase (Days)
	45.58	₹25,039	₹24,666	64.13	24.47

Cluster ID	Age (Years)	Monthly Household Income (₹)	Amount Spent on Previous Phone (₹)	Usage Before Replacing (Months)	Research Before Purchase (Days)
Cluster 0					
Cluster 1	31.75	₹117,309	₹130,088	21.52	13.58
Cluster 2	34.38	₹23,040	₹18,851	65.62	7.74

Distribution of Sample Volume

Cluster ID	Core Persona Classification	Sample Size (Respondents)	Percentage of Sample
Cluster 0	Value-Stretching Loyalists	38	25.5%
Cluster 1	Premium Upgraders	64	43.0%
Cluster 2	Impulsive Budget Buyers	47	31.5%
Total Cohort Pool		149	100.0%

3. Comprehensive Cluster Profiles & Personas

Cluster 0: "Rural Value-Stretching Loyalists" (n = 38)

Demographic Profile: The oldest demographic subset with an average age of approximately 46 years, characterized by a conservative household income profile (~₹25K/month).

Consumer Behavior: They show highly deliberate purchasing tendencies, spending the longest period researching options (~24.4 days). They have a low budget ceiling for tech acquisitions but extract maximum lifespan out of products, holding devices for over 5 years (64.1 months) before upgrading.

Cluster 1: "Urban Premium Upgraders" (n = 64)

Demographic Profile: The largest, younger-to-middle-aged segment (~32 years) possessing an affluent, urban economic profile with monthly household incomes averaging ~₹117K.

Consumer Behavior: This group is the primary premium flagship driver, investing an average of ~₹130K per smartphone purchase. They cycle through hardware quickly, updating their models every 21.5 months while engaging in a moderate two-week window (~13.6 days) of research.

Cluster 2: "Rural Impulsive Budget Buyers" (n = 47)

Demographic Profile: Middle-aged consumers (~34 years) with a constrained economic baseline (~₹23K/month income).

Consumer Behavior: Unlike Cluster 0, this segment does almost no proactive pre-purchase due diligence, executing purchase decisions in roughly 7.7 days. They buy lower-tier devices (~₹18.8K) and hold onto them for an extended duration (~65.6 months), indicating high reliance on in-store advice or immediate convenience deals.

4. Chi-Square Independence Analysis

To evaluate how demographic traits and consumer behaviors interact with our five clusters, we performed Chi-Square tests of independence. A high statistical value accompanied by a low *p*-value indicates that a categorical factor significantly impacts cluster assignment.

Summary Matrix of Categorical Independence Tests

Categorical Variable Tested	Calculated p-value	Statistical Interpretation	Strategic Status
Location Type	< 0.0001	Extremely Significant Correlation	Critical Lever
Current Mobile Phone Brand	0.0230	Statistically Significant Link	Highly Relevant
Gender	0.2380	Not Statistically Significant	Independent

Categorical Variable Tested	Calculated p-value	Statistical Interpretation	Strategic Status
Factors For Buying	0.3830	Not Statistically Significant	Independent
Preferred Purchase Channel	0.8880	Completely Independent	Independent

Detailed Contingency Breakdowns

1. Location Type Analysis (p < 0.0001)

This test delivered the absolute strongest statistical signal in the report. Cluster 1 is comprised 100% of Urban respondents, whereas Clusters 0 and 2 are almost entirely Rural. Even though location was excluded from the algorithm's calculation inputs, the internal socioeconomic metrics perfectly replicated this geographic divide.

Location	Cluster 0 (Rural Loyalists)	Cluster 1 (Urban Premium)	Cluster 2 (Rural Impulsive)
Rural	36	0	47
Urban	2	64	0

2. Current Mobile Phone Brand (p = 0.023)

A clear brand affinity correlation is visible, specifically highlighting ecosystem polarization. 75% of Apple owners reside securely inside Cluster 1 (Premium Upgraders), while Xiaomi over-indexes noticeably inside Cluster 0 (48% of Xiaomi users fall into the Value-Stretching category).

Statistical Note: Due to an 11-brand matrix over 3 distinct groups, roughly 67% of individual data cells possess expected baseline counts below 5. While the primary trajectories for top-market players like Apple and Xiaomi remain highly actionable, the absolute value should be treated as a strong directional trend rather than rigid mathematical certainty.

3. Gender Matrix Breakdown (p = 0.238)

Gender structures stay fairly consistent across the sample pool. Females split roughly 32% / 40% / 28% across groups, while Males report a 20% / 45% / 34% distribution. The variation is small enough to be attributed entirely to random sampling variation, meaning gender does not dictate cluster classification.

Gender	Cluster 0	Cluster 1	Cluster 2
Female	21	26	18
Male	17	38	29

4. Preferred Purchase Channel ($p = 0.888$)

This test approached nearly total independence. Every single cohort reflects almost the exact same 45/55 distribution between brick-and-mortar offline environments and e-commerce platforms. Purchase channel preference does not rely on income tier, geography, or research depth.

Channel	Cluster 0	Cluster 1	Cluster 2
Offline	17	27	22
Online	21	37	25

5. Actionable Marketing Strategies & Conclusions

1. Target Strategy for Cluster 0 ("Value-Stretching Loyalists")

- **Durability Messaging over Specs:** Focus marketing messaging on total cost of ownership, comprehensive multi-year warranties, battery structural integrity, and robust display glass qualities.
- **Long Consideration Funnels:** Given their 24-day research lifecycle, capitalize on early-stage detailed comparison sheets, comprehensive video reviews, and informative localized content.
- **Financial Access Points:** Introduce low-interest or zero-down EMI options and micro-financing to minimize initial barriers to entry.

2. Target Strategy for Cluster 1 ("Premium Upgraders")

- **Structured Trade-in Programs:** Offer subscription upgrade models (e.g., guaranteed trade-in equity at month 18–20) to perfectly fit their fast 22-month upgrade window.
- **Status & Exclusivity Frameworks:** Utilize short-window pre-orders, VIP club memberships, and premium ecosystem bundles (wearables, insurance packages).
- **Urgency Conversion:** Capitalize on short consideration spans (~14 days) by deploying time-sensitive flagship launch promotions.

3. Target Strategy for Cluster 2 ("Impulsive Budget Buyers")

- **In-Store POS & Sales Enablement:** Since these consumers complete research within a week (~8 days), invest heavily in retail promoter visibility, store displays, and dealer recommendation scripts.

- **Frictionless Bundled Value:** Avoid overly detailed technical specifications. Pitch highly transparent values like "Best Battery Under ₹20,000" or simple instant cashbacks.
- **Event-Driven Triggers:** Target this audience at the exact moment of device failure using geo-targeted local promotions and repair network partnerships.